

AgriNova

Crop residue as income. Not smoke.

A working marketplace for farmers, biomass buyers, and agriculture officers in Punjab, with Kisan AI, credits, and a government pollution heatmap.

- Live product: <https://agrinova-ochre.vercel.app>
- Four role-separated desks from a single login
- Punjab case study (Patiala-Sangrur) with lived-in demo data

INSIDE THIS DECK

- 01 Why this product exists
- 02 How the system is built
- 03 User journey (as built)
- 04 Punjab case study
- 05 Product snapshots
- 06 Features in production demo
- 07 Technical impact
- 08 Field and climate impact

The harvest window forces a burn default

After pa
pickup e

Air

Peak November PM2.5 episodes show crop-fire share around 30-40% in public literature. District officers lack a utilised-vs-burned view.

Farmer

Burning earns Rs 0 and risks NGT fines. Smallholders cannot hire a baler alone or reach mill WhatsApp groups.

Buyer

Biomass plants need moisture-qualified straw (often $\leq 15\%$). Informal brokers skip 2-acre farms.

Gap

No shared desk that turns a mapped field into a matched offer, a pickup, and a credit - in one session.

One platform. Four desks. No mixed profiles.

Login se

Farmer

Plot map on OpenStreetMap, residue listing wizard, match score, pickup, credits, Kisan AI, soil card, learn farming.

Buyer

Procurement inbox, moisture specs, RFQs, dual credits, indicative green-tax / CSR simulator.

Government

Department-style portal: pollution heatmap, operational clusters, utilisation slider, savings table.

Admin

Transaction pipeline, carbon ledger, MRV queue for the demo cluster.

Architecture of the running product

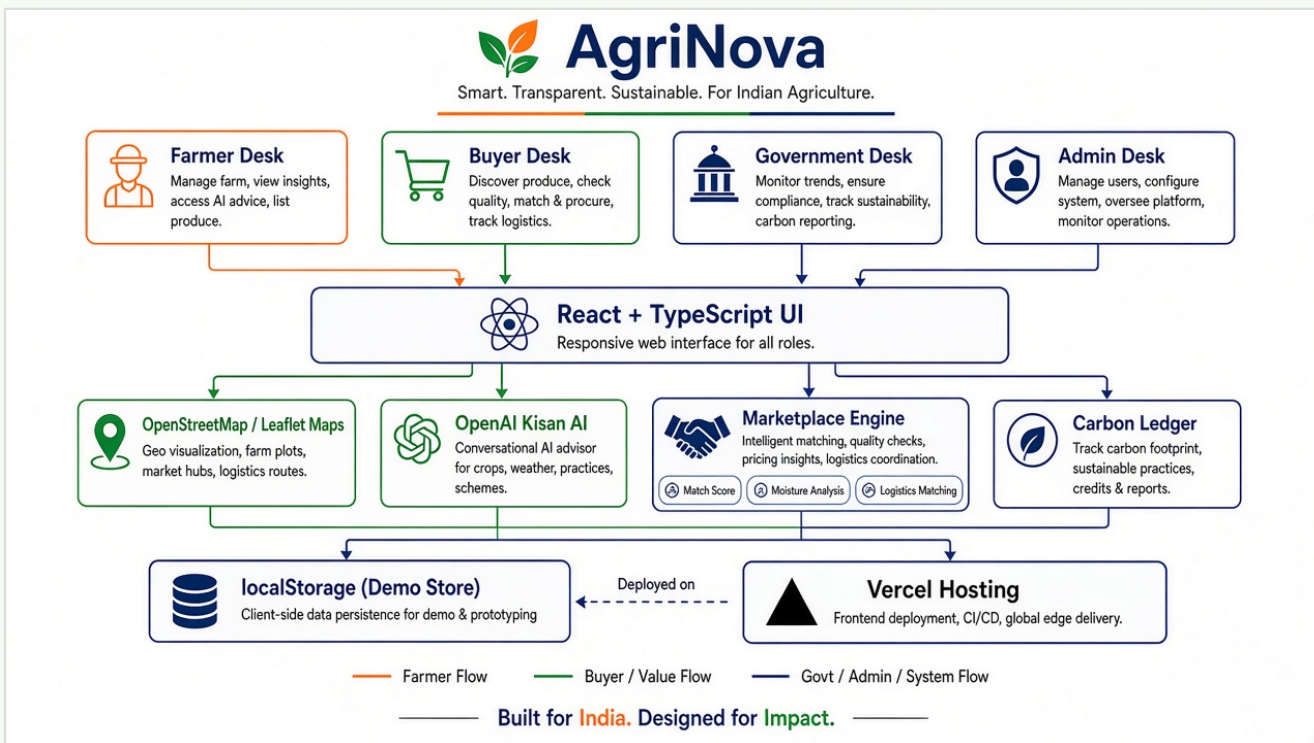


Diagram: four desks, one engine

Roles sit on a React + TypeScript UI. Maps use Leaflet/OSM. Matching is rule-based and moisture-aware. Kisan AI and Policy AI call OpenAI when a key is present, else researched fallbacks. Marketplace state lives in the browser (localStorage) with lived-in seed data so a booth demo is never empty. Hosting is Vercel.

No heavy backend is required for the science-congress booth. Provenance badges mark demonstration data vs public literature vs model estimates.

What the product is built on

Interface	React 19, TypeScript, Vite, Tailwind. White canvas with saffron-white-green tricolor. Role-scoped navigation.
Maps	Leaflet + OpenStreetMap. Farmers draw field corners in lat/long; acres from geodesic area. Government heat and operational markers share the same basemap.
Marketplace	Classify residue, value it, rank pathways, Match Score (distance, demand, moisture, price, pathway). Transaction state machine through pickup and complete.
AI	OpenAI gpt-4o-mini with role system prompts. Voice input (browser speech recognition) and spoken replies. Soil card is injected into farmer context.
Carbon	Avoided tCO2e on completed sales. Farmer wallet (redeem seed/fertiliser). Buyer procurement credits feeding an indicative Rs/tCO2e simulator. Ledger + MRV pages.
Data	Punjab 25-farm pack + West Bengal comparison pack. Lived-in seed: Simran already sold; GreenPower RFQ is open.

From login to credits - and the officer path

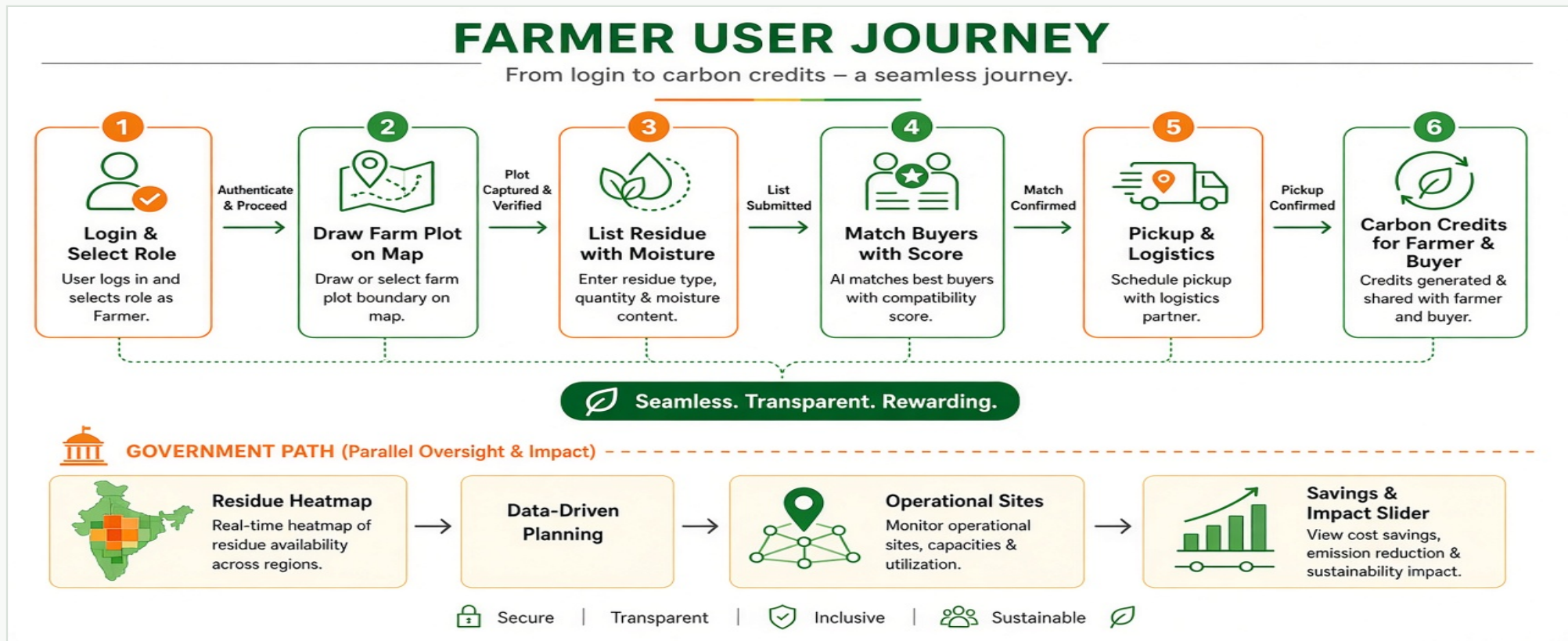


Diagram: farmer path is sequential (role login, map, list, match, pickup, credits). Government path is parallel - heatmap, operational sites, savings slider - with no farmer PII on that desk.

Six steps as implemented

1. Role login

Select Farmer / Buyer / Government / Admin. Confirm name, village, district. One session, one desk.

2. Plot map

Click field corners on OSM around Kharar. Sale parcels estimate straw (~2 t/acre rice). Pathways ranked for that tonnage.

3. List residue

Wizard: farm, quality, moisture, bales, expected Rs/t, pathway. Classification can be overridden.

4. Match

Top buyers scored. Moisture above plant spec is penalised. Quantity and radius remain hard filters.

5. Pickup

Accept offer. Compare individual vs consolidated logistics. Advance the transaction.

6. Credits

On complete: farmer credits for seed/urea; buyer procurement credits for the tax/CSR panel.

Role-first login as built


AgriNova
Login — select your role to continue

Role


Farmer
Sell residue · credits


Buyer
Procure biomass


Government
Air & policy desk


Admin
Operations · MRV

Profile

Ramesh Singh — Smallholder · 6 acres

▼

Name

Ramesh Singh

District

Patiala

Village / office

Kharar

Phone

+91 98765 10421

Land (acres)

6

Age

42

Gender

Male

▼

Login as Ramesh

What this screen does

Four roles on one form. Farmer is selected here with a Punjab profile (Kharar, 6 acres). Login is identity for the demo, not a password IdP. Citizen and officer desks stay separate after this step.

Farmer workspace home


AgriNova
Farmer workspace

Dashboard

Plot map

Sell residue

My listings

Credits

EN

HI

BN

RS

Ramesh Singh

Farmer

↔

SIGNED IN

Ramesh Singh

Kharar, Patiala

Dashboard

Plot map

Sell residue

My listings

Credits

Kisan AI

Learn farming

Pickup

Demand

Weather

Mandi

Ledger

RS

Ramesh Singh

Kharar, Patiala · 6 acres · rice-wheat · KCC ---4821

List residue

"12 days left for wheat. Last year I burned. This year I want the money and no notice from the thana."

Days to wheat sowing

12

PUBLIC DATA

Typical 10-20 day CRM window.

Straw on the field now

5.0 t

2.5 rice acres × ~2 t/acre (IARI-style factor)

Credit wallet

0

Redeem for urea / seed →

This season's job

Sell ~5 t rice straw before the window closes

Last kharif you burned 4.8 t — ₹0 and a fire risk. Moisture, bales, and land area go on the listing so Priya at GreenPower can bid.

Start listing →

Map sown plots

⚠ Rice risk — harvest / residue overlap

Harvest early, list residue within 48hrs, book baler immediately

What this screen does

Signed-in farmer only. Sidebar is the residue desk: map, sell, listings, credits, Kisan AI, learn, pickup. Language toggle EN/HI/BN. Switch account is logout - not a mid-session role flip.

Real map, drawn parcels, straw estimate

AgriNova
Farmer workspace

Dashboard Plot map Sell residue My listings Credits

EN HI BN

RS Ramesh Singh
Farmer

SIGNED IN
Ramesh Singh
Kharar, Patiala

Dashboard

Plot map

Sell residue

My listings

Credits

Kisan AI

Learn farming

Pickup

Demand

Weather

Mandi

Ledger

Map your sown plots

Draw sown fields on the real OpenStreetMap around Kharar. Mark parcels for sale — AgriNova estimates straw and ranks pathways.

Crop for next draw

Rice ☒ Mark parcel for sale Clear stroke Save parcel Reset parcels

OpenStreetMap - click field corners on the real map around Kharar / Patiala. Acres from geodesic area.

Paddy block A: 2.5 ac · Rice · for sale · ~5.0 t (30.7389, 76.6486)

Best use pathways for ~5.0 t

Biomass power / industry Strong buyer demand and established haul routes.	92
Biochar production Higher price per tonne; suitable for dry straw.	82
Composting	74

What this screen does

OpenStreetMap around Kharar. Green polygon is a sale parcel. 2.5 acres rice maps to ~5.0 t straw. Pathways (biomass 92, biochar 82, compost 74) sit under the map so geography feeds the market.

List residue: farm, quality, match

AgriNova

Farmer workspace

Dashboard

Plot map

Sell residue

My listings

Credits

SIGNED IN

Ramesh Singh

Kharar, Patiala

Dashboard

Plot map

Sell residue

My listings

Credits

Kisan AI

Learn farming

Pickup

Demand

Weather

Mandi

Ledger

EN

HI

BN

RS

Ramesh Singh

Farmer

[→]

List residue — Ramesh Singh

2.5 acres paddy × ~2 t/acre ≈ 5.0 t straw. Moisture and bales decide whether GreenPower can lift it.

1. farm

2. quality

3. listing

4. pathways

5. offers

6. logistics

7. complete

Farm & land

Farmer

Ramesh Singh

Phone

+91 98760000

Location

Kharar, Patiala, Punjab

Total land (acres)

6

Area under this residue (acres)

5

Source crop

Rice

Harvest date

01-11-2025

Detected residue

AI OUTPUT

Rice Straw

96% confidence

Rice Straw

Wheat Stubble

Cotton Stalks

Sugarcane Tops

Maize Stover

Continue

What this screen does

Seven-step golden path. AI classification of Rice Straw with override chips. Moisture later gates whether GreenPower can lift the lot. This is the 5-minute judge path from field to offer.

AgriNova | Product as built | NCSC

11/22

Soil monitoring card + OpenAI chat


AgriNova
Farmer workspace

[Dashboard](#)
[Plot map](#)
[Sell residue](#)
[My listings](#)
[Credits](#)

EN HI BN


Ramesh Singh
Farmer

[->]

SIGNED IN

Ramesh Singh

Kharar, Patiala

[Dashboard](#)
[Plot map](#)
[Sell residue](#)
[My listings](#)
[Credits](#)
[Kisan AI](#)
[Learn farming](#)
[Pickup](#)
[Demand](#)
[Weather](#)
[Mandi](#)
[Ledger](#)

Kisan AI
 Tuned for residue, soil, and sowing — uses your soil card and plot map.

Soil profile

Soil type

Alluvial

pH

7.2

Organic carbon %

0.45

Nitrogen status

Medium

Notes for Kisan AI

Patiala belt alluvial — demo soil card for Kisan AI.


KisanSathi
Farmer Ramesh Singh, Kharar, Patiala. Land 6 ac. Soil alluvial pH 7.2. Mapped sa...

🔊

OpenAI live


 Type or tap the mic — Hindi/English. Answers use OpenAI when configured.

What this screen does

Editable soil card: alluvial, pH 7.2, OC 0.45%, N medium. KisanSathi uses that card plus mapped acres. Mic and spoken replies. OpenAI live when a local key is set; otherwise researched fallbacks.

New methods taught inside the same desk

SIGNED IN

Ramesh Singh

Kharar, Patiala

Dashboard

Plot map

Sell residue

My listings

Credits

Kisan AI

Learn farming

Pickup

Demand

Weather

Mandi

Ledger

Dashboard

Plot map

Sell residue

My listings

Credits

EN HI BN

RS Ramesh Singh Farmer

[→]

Kisan learning

New farming methods

Pick a practice, then ask Kisan AI by text or voice (OpenAI). Hindi + English ok.

All

Residue

Sowing

Soil

Water

Income

Happy Seeder / zero-till wheat

हैप्पी सीडर

Sow wheat into standing rice residue without burning. Saves time between harvest and sowing.

Direct Seeded Rice (DSR)

डीएसआर धान

Rice without puddling transplant. Cuts labour and can save irrigation water when timed well.

Bale & sell residue

पराली बेचना

Bale dry straw ($\leq 15\%$ moisture) and sell via AgriNova instead of burning.

On-farm biochar pathway

बायोचार

Convert part of residue to biochar for soil carbon — slower cash than biomass sale.

Surface mulch & incorporation

मलच / मिट्टी में मिलाना

Leave chopped residue as mulch or incorporate lightly where buyers are far.

Pulse intercrop / diversification

दलहन अंतरफसल

Add short-duration pulses or oilseeds where water and market allow.

Happy Seeder / zero-till wheat

- Avoids residue burn fines
- Keeps soil cover
- Often lower diesel than full tillage

Voice on in chat

Learn: Happy Seeder / zero-till wheat

Ramesh Singh · Patiala

OpenAI live

Type or tap the mic — Hindi/English. Answers use OpenAI when configured.

What this screen does

Practice cards (Happy Seeder, DSR, biochar, mulch, pulses) seed the chatbot. Voice on. The farmer stays in-role - learning is not a separate government portal.

AgriNova | Product as built | NCSC

13/22

Farmer wallet after sales

 **AgriNova**
Farmer workspace

Dashboard Plot map Sell residue My listings **Credits**

EN HI BN RS **Ramesh Singh**
Farmer

SIGNED IN
Ramesh Singh
Kharar, Patiala

Dashboard Plot map Sell residue My listings **Credits** Kisan AI Learn farming Pickup Demand Weather Mandi Ledger

Carbon credit wallet

Ramesh starts at 0. Simran Kaur (Sangrur) already earned credits from 3.2 t stubble and redeemed urea — check Ops / ledger.

AVAILABLE BALANCE

 0

Lifetime earned: 0
Redeemed: 0
From ledger: 0 cr.

AgriNova Credits · 1 credit = 1 kg CO₂e avoided (demo unit)

Redeem for farm inputs

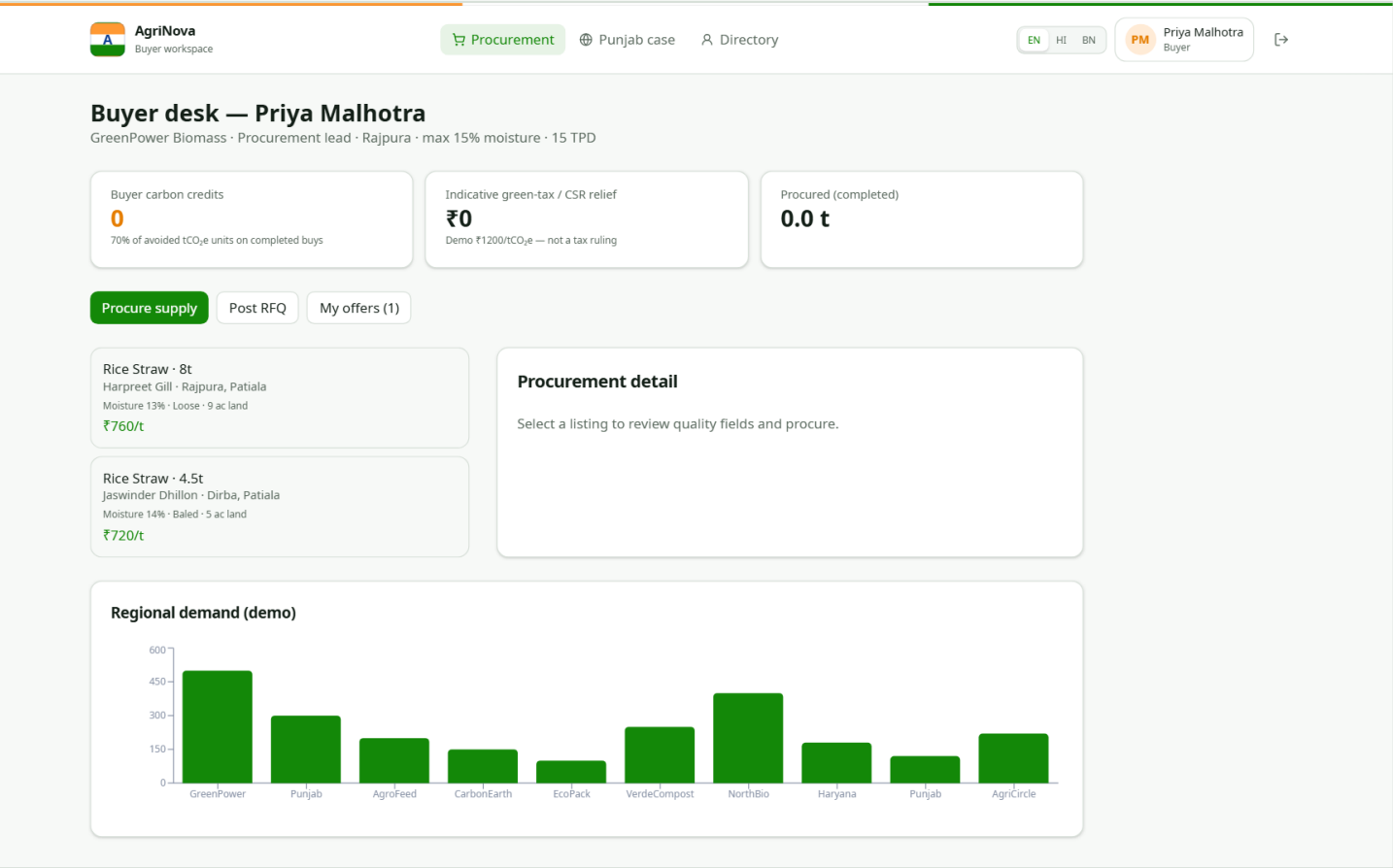
Wheat seed pack (5 kg) 120 cr PBW-343 variety — demo subsidy via credits Redeem	Paddy seed (2 kg) 95 cr Short-duration variety for timely sowing Redeem
Urea (1 bag) 180 cr Redeem at partner agro-dealer (demo) Redeem	DAP (1 bag) 220 cr Phosphorus for Rabi crops Redeem
Baler rental (1 day) 350 cr Shared baler in Patiala cluster Redeem	Soil health test 80 cr Lab kit via Krishi Vigyan Kendra partner Redeem

 Completing a sale on the marketplace issues credits automatically. This prototype is not a regulated

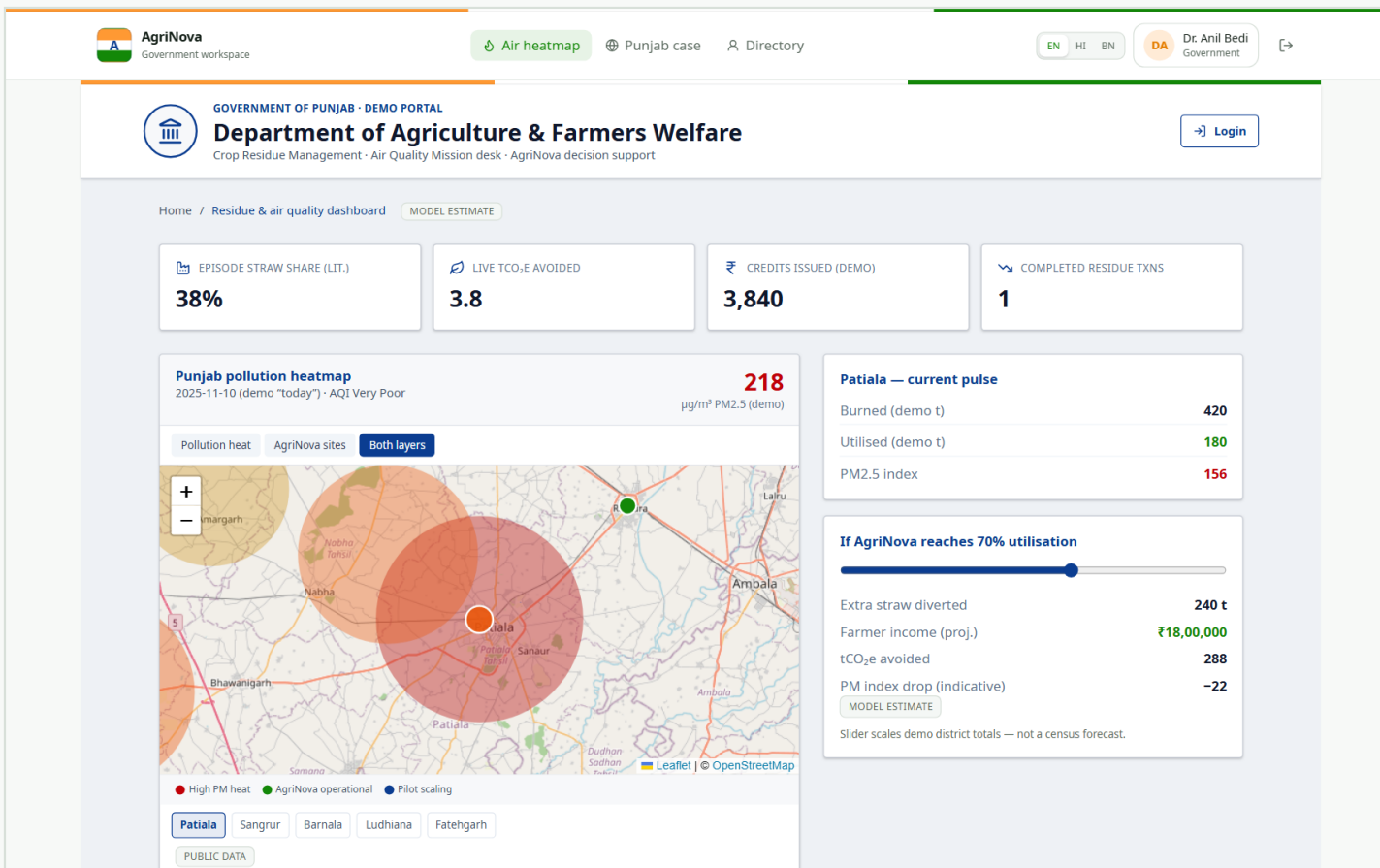
What this screen does

Credits from completed residue sales. Redeem against seed/fertiliser catalogue. Simran's lived-in sale is already in seed data so this screen is never empty on first load.

Procurement desk as built



Agriculture portal: heatmap + savings



What this screen does

Department of Agriculture masthead. Heat blobs = demo PM intensity. Green markers = operational AgriNova clusters. Patiala pulse + utilisation slider (70% -> extra 240 t, Rs 18 lakh farmer income, 288 tCO₂e). Model estimate, not a census forecast.

Patiala-Sangrur cluster as the product's home

The run
practice

Cluster

25 farms, ~180 t straw in the pack. Default story:
2.5 acres paddy x ~2 t/acre = ~5 t for one listing.

Places

Kharar (farmer plots), Rajpura (GreenPower plant, 8 km), Sangrur (second farmer), Patiala DAO desk.

Without AgriNova

148 t burned / 32 t utilised in the pack. Farmer income Rs 0. Indicative PM index 172.

With 70% utilisation

54 t burned / 126 t utilised. Farmer income Rs 9.45 lakh. ~151 tCO₂e avoided. PM index 138.

Operational sites

Kharar, Sunam, Rajpura live; Barnala pilot. Rollup: straw diverted, burn avoided, income, tCO₂e on the map.

Honesty layer

Labels: demonstration data, model estimate, public data. No statewide forecast from 25 farms.

Capability list in the live product

Role login

Farmer, buyer, government, admin - separate nav, RoleGate on URLs.

OSM plot map

Draw parcels, geodesic acres, straw estimate, pathway ranking.

Sell wizard

Classify, value, moisture, offers, logistics, complete.

Match Score

Five factors + moisture penalty vs plant spec.

Kisan AI

OpenAI, Hindi/English, soil-aware, voice in and out.

Soil card

Type, pH, OC%, N status persisted in session.

Learn farming

Happy Seeder, DSR, biochar, mulch, diversification.

Dual credits

Farmer redeem + buyer procurement / CSR panel.

Gov heatmap

Heat + operational markers + utilisation slider.

Carbon ledger

Avoided tCO2e entries and MRV queue.

Lived-in seed

Prior sale and open RFQ so tables are not empty.

i18n

EN / HI / BN toggle on the shell.

What the engineering choices enable

Explainable matching

Judges can open a score breakdown: distance, demand, moisture, price, pathway. Not a black box.

Geography is real

OSM tiles and lat/lng parcels make straw-from-land credible. Heatmap uses district HQ coords, not a fake sketch.

AI is bounded

Role system prompts. Soil and acres injected. Missing key -> fallback copy so the booth still runs.

State is demo-safe

localStorage + seed data. Reset possible. No farmer PII on the government desk.

Honest models

Utilisation slider and before/after charts are labelled MODEL ESTIMATE and scaled to the cluster.

Deployed

SPA on Vercel with SPA rewrites. Live URL for judges who will not install Node.

If straw is sold instead of burned

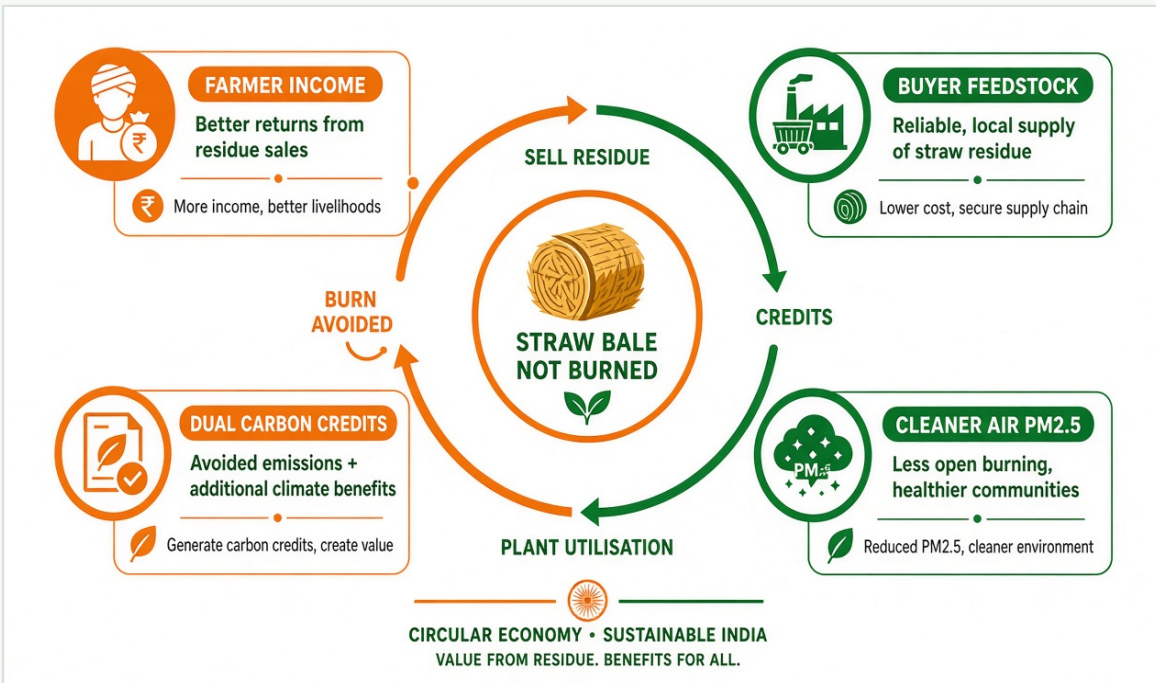


Diagram: circular residue loop

Burn avoided feeds utilisation. Plants get feedstock. Farmers get Rs/t plus redeemable credits. Buyers get procurement credits. District PM index falls in the model. The loop is the product: marketplace + credits + officer view.

- Farmer: cash vs Rs 0 burn; credits for seed/fertiliser; no mid-window thana notice if residue is lifted.
- Buyer: moisture-qualified lots from farms that never reached mill groups.
- Officer: utilised vs burned by district, operational cluster savings, dual-credit totals - aggregates only.
- Climate: avoided tCO2e on completed transactions; 70% utilisation pack ~151 tCO2e in the demo model.

Question the product is designed to test

Can an

Hypothesis

Income + moisture-aware match + pickup logistics + soil-aware advice beats the burn habit for a 4-6 acre rice-wheat farm.

Method

Interactive simulation. Lived-in seed. Complete a sale in-session. Compare without vs 70% utilisation. West Bengal pack for comparative CRM.

Limits (stated in the UI)

Not cadastral land records. Not live CPCB sensors. Not a certified carbon registry. Buyer tax panel is indicative, not a finance-ministry rule. Cluster numbers are not state totals.

AgriNova is live

Open th

<https://agrinova-ochre.vercel.app>

Login -> Farmer -> Plot map -> Sell residue

Login -> Government -> Air heatmap + savings slider

Login -> Farmer -> Kisan AI (soil + voice)

In-app slides: /presentation

Tagline

Predict.

Protect.

Prosper.

Recycle.

National Children's Science Congress | Product as built | Punjab case study demo